

Efficient Skeleton-Based Action Recognition using Superposed Shape Subspace

anonymous

CONFIDENTIAL. For review only.

Abstract

We propose a lightweight and generalizable method for skeleton-based action recognition using a novel representation called the *superposed shape subspace* (SSS). Our method captures the structural and temporal dynamics of skeleton sequences by representing multiple frames as a single subspace in a high-dimensional space. Unlike deep learning models that require significant training data and GPU resources, our approach performs efficient classification using canonical angles between subspaces, operating entirely on CPU with rapid training and inference. We further introduce an optimal frame selection mechanism that reduces computational cost by focusing only on key informative frames. Experiments on the First-Person Hand Action Benchmark demonstrate that our method achieves competitive accuracy (89%) while requiring only a fraction of the processing time and memory, and can adapt to new classes with minimal data. This makes it suitable for edge computing and real-time applications.

1 Introduction

Skeleton-based action recognition is fundamental for human-computer interaction in AR/VR, robotics, and accessibility applications [9, ?, 10]. While deep learning approaches have achieved high accuracy, they typically demand extensive computational resources, long training times, and large datasets.

This paper presents a method for action recognition based on shape representation that offers three distinct advantages: (1) it processes skeletal data at over 100 fps on standard CPU hardware, (2) it can be trained in seconds rather than hours, (3) and it enables one-shot learning of new action classes, meaning the system can recognize a new type of action after seeing just a single example video. Our approach models the structure of 3D skeleton data as a three-dimensional subspace, called the shape subspace, within a high-dimensional vector space. The similarity between two skeletons is assessed through the canonical angles between their corresponding shape subspaces [1, 3].

Our key innovation is the superposed shape subspace (SSS), which integrates multiple skeletal poses across sequential frames into a unified representation.

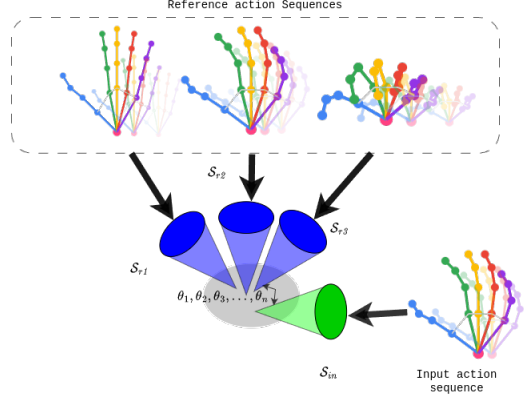


Figure 1. Action recognition using superposed shape subspace representation. Each subspace is generated from consecutive keyframes of skeleton data, capturing both spatial structure and temporal dynamics.

This captures both spatial structure and temporal dynamics in a single mathematical framework. However, the computational requirements of this approach can be substantial. To address this challenge, we implement an optimal frame selection mechanism that intelligently selects the most informative frames, significantly reducing computational costs while maintaining accuracy.

Experiments on the First-Person Hand Action Benchmark dataset [12] demonstrate that our method achieves 89% accuracy across 45 action classes while processing at 108 frames per second on standard CPU hardware. While deep learning approaches may achieve marginally higher accuracy (typically 2-7% higher), our method offers dramatic advantages in computational efficiency, training speed, and adaptability—making it ideal for resource-constrained real-time applications.

Our primary contributions are as follows. First, we introduce a novel superposed shape subspace representation that effectively captures temporal dynamics by

integrating multiple frames before subspace generation. Second, we develop an optimal frame selection mechanism that reduces computational requirements by 58% while maintaining high recognition accuracy. Third, we present a computationally efficient approach that achieves competitive accuracy (89%) while processing at 108 fps and enabling one-shot learning of new action classes.

2 Theoretical Fundamentals

2.1 Shape Subspace Representation

A shape subspace is a compact representation of the geometrical structure of a set of feature points [5, 6]. For hand gesture recognition, we represent a hand pose as a 3-dimensional shape subspace in a high-dimensional vector space, where the dimensionality is determined by the number n_l of joint landmarks of the hand.

Given a hand pose frame $F \in \mathbb{R}^{n_l \times 3}$ containing n_l landmarks, we generate a shape subspace through:

1. Centering the points to achieve translation invariance:

$$\hat{F} = F - \frac{1}{n_l} \sum_{i=1}^{n_l} F_i, \quad (1)$$

2. Computing SVD to find principal directions:

$$\hat{F} = U \Sigma V^T, \quad S = U_{:,1:3}, \quad (2)$$

The resulting 3-dimensional subspace S captures the essential structure of the skeletal pose while maintaining invariance to translation and rotation—a key advantage over raw coordinate-based approaches.

2.2 Strutal similarity between two shape subspaces

We measure the strutal similarity between two shape subspaces by the canonical angles between them. Given three-dimensional shape subspaces P and Q in the n_l -dimensional vector space, the canonical angles $\{0 \leq \theta_1 \leq \theta_2 \leq \theta_3 \leq \pi/2\}$ between P and Q are uniquely defined as:

$$\cos^2 \theta_i = \max_{\substack{u_i \perp u_j (i \neq j, i, j=1 \sim 3) \\ v_i \perp v_j (i \neq j, i, j=1 \sim 3)}} \frac{(u_i \cdot v_i)^2}{\|u_i\|^2 \|v_i\|^2}, \quad (3)$$

where $u_i \in P$, $v_i \in Q$, $u_i \neq 0$, $v_i \neq 0$.

The similarity between the shape subspaces is defined as:

$$S = \frac{1}{3} \sum_{i=1}^3 \cos^2 \theta_i. \quad (4)$$

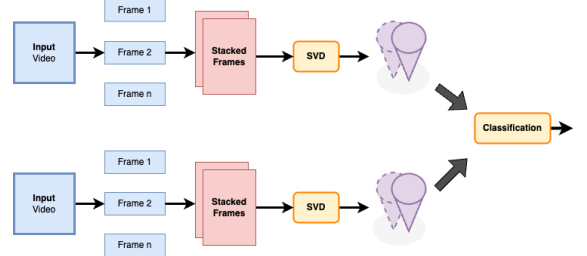


Figure 2. Overview of our superposed shape subspace method with optimal frame selection.

3 Proposed method

3.1 Superposed Shape Subspace (SSS)

To effectively capture the temporal dynamics inherent in actions like hand gestures, we introduce the *superposed shape subspace (SSS)*. While the basic shape subspace described previously in 2.1 represents the geometry of a single skeleton frame, the SSS extends this concept to represent the combined structure across multiple consecutive frames within a temporal window. This allows the representation to encode not just the pose at one instant, but also the motion occurring over a short time period.

The generation of an SSS involves integrating information from a sequence of frames before computing the subspace basis. Given a sequence of hand skeleton data, we use a sliding window approach. Let w be the size of the temporal window (i.e., the number of consecutive frames to consider). The SSS for a window starting at time t is computed as follows:

1. **Window Selection:** Select w consecutive hand pose frames from the sequence: $\{F_t, F_{t+1}, \dots, F_{t+w-1}\}$. Each frame $F_i \in \mathbb{R}^{n_l \times 3}$ contains the 3D coordinates for n_l landmarks.
2. **Individual Frame Centering:** Center each frame F_i within the window independently to achieve translation invariance for each pose:

$$\hat{F}_i = F_i - \frac{1}{n_l} \sum_{j=1}^{n_l} (F_i)_j \quad \text{for } i = t, \dots, t+w-1, \quad (5)$$

where $(F_i)_j$ denotes the 3D coordinates of the j -th landmark in frame i .

3. **Stacking Centered Frames:** Vertically stack the w centered frames into a single combined matrix:

$$\hat{F}_{window} = \begin{bmatrix} \hat{F}_t \\ \hat{F}_{t+1} \\ \vdots \\ \hat{F}_{t+w-1} \end{bmatrix}. \quad (6)$$

This matrix $\hat{F}_{window}$ has dimensions $(w \cdot n_l) \times 3$ and encapsulates the spatial configuration of landmarks from all w frames in a single structure.

4. **SSS Basis Computation via SVD:** Compute the SVD of the combined matrix:

$$\hat{F}_{window} = U\Sigma V^T, \quad S_{SSS} = U_{:,1:3}. \quad (7)$$

The resulting S_{SSS} is a 3-dimensional subspace embedded within the $(w \cdot n_l)$ -dimensional space, capturing the principal geometric variations across the w frames.

By applying this process with a sliding window across the entire action sequence, a full gesture is transformed into a sequence of superposed shape subspaces $\{S_1, S_2, \dots, S_m\}$. This approach ensures that both the static shape of the hand and its dynamic evolution over time are encoded in the representation.

3.2 Optimal Frame Selection

Generating Superposed Shape Subspaces (SSS) for densely overlapping windows across an entire video sequence can be computationally intensive. To improve efficiency and focus computational resources on the most informative parts of the action, we employ frame selection techniques before generating the SSS sequence. These techniques aim to create a sparser, yet representative, sequence of frames from the original video. We implement two primary strategies:

1. **Temporal Downsampling:** This technique uniformly reduces the number of frames processed. We select frames from the original sequence at regular intervals, effectively skipping frames. Given an original frame sequence $\{F_1, F_2, F_3, \dots\}$, we create a downsampled sequence by selecting every s -th frame:

$$F_{sampled} = \{F_1, F_{1+s}, F_{1+2s}, \dots, F_{1+k \cdot s}\}, \quad (8)$$

Here, s is the sampling stride (or step size), a positive integer greater than 1 (e.g., $s = 2$ means selecting every second frame). N is the total number of frames in the original sequence. This method significantly reduces the number of frames to consider for SSS generation.

2. **Keyframe Detection based on Motion:** Instead of uniform sampling, this technique adaptively selects frames where significant motion occurs, assuming these frames are more critical for representing the action dynamics. We quantify the amount of motion between consecutive frames F_{i-1} and F_i by calculating the average Euclidean distance change across all n_l landmarks:

$$diff_i = \frac{1}{n_l} \sum_{j=1}^{n_l} \|F_i(j) - F_{i-1}(j)\|_2, \quad (9)$$

where $F_i(j)$ denotes the 3D coordinates of the j -th landmark in frame i . A frame F_i is selected as a

keyframe if its associated motion $diff_i$ exceeds a predefined motion threshold τ :

$$\text{Select } F_i \text{ if } diff_i > \tau. \quad (10)$$

These frame selection techniques produce a reduced set of frames. The SSS generation process is then applied to this reduced set, using sliding windows over the selected frames rather than the original dense sequence. This significantly lowers the computational cost while aiming to preserve high recognition accuracy by retaining essential temporal information.

3.3 Classification process

Once an input action video is processed and represented as a sequence of Superposed Shape Subspaces (SSS) $\{S_1, S_2, \dots, S_m\}$ (as detailed in Section 3.1), the next step is to classify this sequence to identify the action performed. Our classification approach leverages the structural similarity between subspaces and employs a sequence-level voting scheme for robustness. This overall process is depicted in Figure 2. The classification proceeds as follows:

1. **Reference Database:** We assume a pre-computed reference database exists, containing labeled SSS instances derived from training video sequences representing the different action classes. Let this database contain reference subspaces R_k , each associated with a known action class label $C(R_k)$.

2. **Per-Subspace Similarity Calculation:** For each SSS, S_i , in the input sequence $\{S_1, \dots, S_m\}$, we compute its structural similarity to every reference subspace R_k in the database. This similarity is measured using the canonical angles between the two subspaces, as defined in Section 2.2. A higher value indicates greater similarity.

3. **Per-Subspace Classification (k-NN):** Each input subspace S_i is individually classified using a k-Nearest Neighbors (k-NN) algorithm based on the similarity scores calculated in the previous step. We identify the k reference subspaces $\{R_{k_1}, R_{k_2}, \dots, R_{k_k}\}$ from the database that are most similar to S_i . The class label $C(S_i)$ assigned to the input subspace S_i is determined by the majority class among these k nearest neighbors:

$$C(S_i) = \underset{c}{\operatorname{argmax}} \sum_{j=1}^k [C(R_{k_j}) = c], \quad (11)$$

where $[\cdot]$ is the Iverson bracket (evaluating to 1 if the condition inside is true, and 0 otherwise).

4. **Final Action Classification (Majority Voting):** After classifying each SSS (S_i) in the input sequence, we obtain a sequence of class labels $\{C(S_1), C(S_2), \dots, C(S_m)\}$. The final classification for the entire action gesture, $C_{gesture}$, is determined by

Table 1. Comparison with other methods.

Method	Accuracy
DD-net [7]	0.88
LSTM-based [8]	0.84
Basic Subspace	0.78
Our Method	0.89

performing a majority vote over this sequence of individual subspace classifications:

$$C_{gesture} = \underset{c}{\operatorname{argmax}} \sum_{i=1}^m [C(S_i) = c]. \quad (12)$$

The action class c that receives the most votes across the entire sequence of SSS is assigned as the final recognized action.

This "sequence-based" classification approach, by aggregating decisions from multiple subspaces representing different temporal segments of the action, enhances the robustness of the recognition. It makes the final classification less sensitive to potential misclassifications of individual subspaces that might occur due to noise, variations in execution speed, or ambiguities in specific poses within the action sequence.

4 Experimental Results

4.1 Dataset and Performance Evaluation

We evaluated our superposed shape subspace method using the First-Person Hand Action Benchmark dataset, which consists of 45 different daily hand action categories performed by 6 different subjects, involving interactions with 26 different objects.

For evaluation, we considered both binary classification (between two similar actions) and full 45-class classification. Our method achieved 94% accuracy in binary classification and 89% in the full 45-class task. Computationally, our approach processes only 42% of the frames compared to conventional methods, with a processing time of 9.2 ms/frame and minimal memory overhead ($1.08\times$ baseline).

As shown in Table 1, our method achieves competitive performance compared to other state-of-the-art approaches, slightly outperforming DD-net (89% vs 88%) and significantly outperforming LSTM-based methods (89% vs 84%). While the standard superposed subspace method achieves slightly higher accuracy (93%), our method requires less than half the computational resources, making it more suitable for practical applications.

5 Conclusion

We have presented an enhanced superposed shape subspace method with optimal frame selection for ac-

tion recognition. Our approach effectively captures temporal dynamics by stacking consecutive frames before subspace generation and implements intelligent frame selection to significantly reduce computational requirements. Experimental results demonstrate 89% accuracy across 45 action classes while processing less than half the frames required by conventional approaches.

The key innovation is the effective integration of temporal information through frame stacking combined with intelligent frame selection that preserves only the most informative frames. This approach balances recognition accuracy and computational efficiency, making it suitable for resource-constrained real-time applications. Future work will explore adaptive window sizing and their application to additional skeletal datasets.

References

- [1] Fukui, K.: "Subspace Methods," *Computer Vision*, pp.777–781, 2014.
- [2] Fukui, K., Maki, A.: "Difference Subspace and Its Generalization for Subspace-Based Methods," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol.37, no.11, pp.2164–2177, 2015.
- [3] Maeda, K., Watanabe, S.: "A Pattern Matching Method with Local Structure," *Transactions of IEICE*, vol.J68-D, no.3, pp.345–352, 1985.
- [4] Fukui, K., Yamaguchi, O.: "Face Recognition Using Multi-viewpoint Patterns for Robot Vision," *11th International Symposium of Robotics Research*, pp.192–201, 2003.
- [5] Igarashi, Y., Fukui, K.: "3D Object Recognition Based on Canonical Angles Between Shape Subspaces," *Proceedings of the 10th Asian Conference on Computer Vision*, pp.580–591, 2010.
- [6] Yoshinuma, T., Hino, H., Fukui, K.: "Personal Authentication Based on 3D Configuration of Micro-feature Points on Facial Surface," *Image and Video Technology - 7th Pacific-Rim Symposium*, pp.433–446, 2015.
- [7] Yataka, R., Fukui, K.: "Three-dimensional Object Recognition via Subspace Representation on a Grassmann Manifold," *Computer Vision and Applications*, vol.33, no.2, pp.40–49, 2022.
- [8] Fukui, K., Valois, P.H.V., Souza, L., Kobayashi, T.: "Second-order difference subspace," *arXiv preprint arXiv:2409.08506*, 2024.
- [9] Rautaray, S. S., Agrawal, A.: "Vision based hand gesture recognition for human computer interaction: a survey," *Artificial Intelligence Review*, vol.43, no.1, pp.1–54, 2015.
- [10] Chen, M., AlRegib, G., Juang, B. H.: "Air-writing recognition—Part I: Modeling and recognition of characters, words, and connecting motions," *IEEE Transactions on Human-Machine Systems*, vol.46, no.3, pp.403–413, 2013.
- [11] Papadopoulos, G. T., Tzovaras, D., Strintzis, M. G.:

“Adaptive keyframe extraction using visual attention for video summarization,” *IEEE International Conference on Image Processing (ICIP)*, pp.4104–4108, 2016.

[12] Garcia-Hernando, G., Yuan, S., Baek, S., Kim, T.

K.: “First-person hand action benchmark with RGB-D videos and 3D hand pose annotations,” *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pp.409–419, 2018.